Biology Trees and wood formation

Background guide BG01 1/6

Used mainly in Lessons 2, 3 and 4.

Cells, tissues and organs

Cells form tissues, and organs such as roots, stems and leaves work together. Wood is secondary xylem and represents one part of a tree.

Sites of growth

Apical meristems contribute to primary growth in length. The vascular cambium produces secondary xylem inward and secondary phloem outward. Seasonal changes in wood formation contribute to growth-ring structure.

Cell functions

Tracheids conduct water and support conifer wood. Compare vessels, fibres and parenchyma in hardwoods. Mature conducting cells are generally dead, while sapwood also contains living parenchyma.

Check your understanding

On which side of the bark does wood accumulate? Name the tissues produced inward and outward by the vascular cambium.

Answer Wood accumulates inside the bark. The vascular cambium produces secondary xylem inward and secondary phloem outward.

References

R15 The plant body OpenStax Biology 2e

R17 Stems and plant growth OpenStax Biology 2e

R10 Wood anatomy databases 森林総合研究所 FFPRI

Physiology and ecology Environment and growth

Background guide BG01 2/6

Used mainly in Lessons 2, 3, 6 and 7.

Photosynthesis and carbon allocation

Carbon acquired by photosynthesis supports respiration, leaves, roots, storage and wood formation. Ring width alone does not directly measure total photosynthesis in a year.

Water relations and phenology

Xylem transport, stomata and transpiration are linked. Temperature, water availability, leaf-out and dormancy influence growth. Verify seasonality for a species and site before treating growth rings as annual.

Individuals and forests

Competition for light, water and nutrients, canopy gaps, insects and fire can alter growth. Inference from one tree to a population or forest requires replication, spatial information and additional evidence.

Check your understanding

Rings widen after a neighbouring tree falls. How could this be distinguished from improved climatic conditions?

Interpretation Compare nearby and distant trees, canopy changes, felling or windthrow records, and climate observations.

References

R16 Transport of water and solutes in plants OpenStax Biology 2e

R17 Stems and plant growth OpenStax Biology 2e

R09 Glossary of Dendrochronology WSL

Mathematics Graphs, ratios and time

Background guide BG01 3/6

Used mainly in Lessons 1, 3, 4 and 7.

Graph axes

First check whether the horizontal axis shows calendar year or tree age, and whether the vertical axis shows mm or an index. Axis ranges and logarithmic scales change the visual impression.

Ratios and indices

If ring width is 1.5 mm and an estimated growth trend is 1.2 mm, a ratio index is 1.5 / 1.2 = 1.25. This is 1.25 times the expected value, not 1.25 mm. These teaching values do not directly measure temperature.

Time intervals

The interval from 1200 CE to 1250 CE is 50 years, but counting both endpoint rings gives 51 rings. Distinguish elapsed time from the number of included years. In the historical calendar, 1 BCE is followed by 1 CE with no year zero.

Check your understanding

What is the ratio index for a ring width of 2.4 mm and an expected value of 2.0 mm? How many rings are included from 1300 to 1320 CE, inclusive?

Answers Index 1.2. Twenty-one rings.

References

R09 Glossary of Dendrochronology WSL

R13 Engineering Statistics Handbook NIST/SEMATECH

Statistics Variation and relationships

Background guide BG01 4/6

Used mainly in Lessons 3, 4, 6 and 7.

Mean and variability

The mean of 1.0, 2.0 and 3.0 mm is 2.0 mm. The values 1.9, 2.0 and 2.1 mm have the same mean but less variability. Examine individual points, sample size and spread.

Correlation and causation

Two variables changing together do not necessarily have a causal relationship. Consider common causes, trends and other influences. Inspect a scatter plot before interpreting a correlation coefficient.

Uncertainty and independent validation

Measurement uncertainty, dating uncertainty and environmental reconstruction uncertainty are different. Climate models need checking beyond agreement with their fitting period, using separate periods or independent records.

Check your understanding

Ten trees show growth reductions in the same year. What additional evidence would you seek before attributing the change to drought?

Possible checks Climate observations, other sites, insect or fire evidence, management history and sampling design. Synchronous change is one line of evidence.

References

R13 Engineering Statistics Handbook NIST/SEMATECH

R08 International Tree-Ring Data Bank NOAA NCEI

R14 Correlation and causality NIST/SEMATECH

Earth science Strata and environmental archives

Background guide BG01 5/6

Used mainly in Lessons 1, 6, 7 and 8.

Stratigraphy

Use superposition and cross-cutting relationships to examine the sequence of formation. In a simple depositional sequence, lower strata are older; check for folding, faulting, erosion and redeposition.

Deposition and preservation

Material can form before transport and deposition. A wood fragment within a layer need not have the same age as deposition of that layer.

Proxies and age models

Separate interpretation of environmental information from dating each layer. Verify that proposed varves are annual. Distinguish temporal resolution from age uncertainty.

Check your understanding

Can the deposition age of a sand layer be established from a radiocarbon date on an old wood fragment within it alone?

Interpretation Assess pre-depositional age and redeposition. Consider short-lived samples or another method addressing deposition.

References

R01 Introduction to dating Smithsonian Human Origins Program

R05 Scientific dating methods Historic England

R07 What are proxy data NOAA NCEI

History and archaeology Linking samples and events

Background guide BG01 6/6

Used mainly in Lessons 1, 5 and 8.

Context

Record where and how a sample was recovered. Species, tool marks, timber position and associated finds help relate a date to an event.

Source criticism

Consider a document's date, author, purpose and transmission. Distinguish a reported event from the date of a copy and later interpretation. Scientific dates likewise need sampling and analytical context.

Ring formation and use

Ring formation can precede felling, storage, manufacture, repair and reuse. If the final outer ring is missing, the youngest surviving ring alone does not establish an exact felling year.

Check your understanding

The youngest surviving ring in a post dates to 1500 CE. What evidence is needed before assigning construction to 1500?

Interpretation Check preservation to the final ring, felling and storage, reuse, other timbers, building evidence and documents. One surviving ring date does not establish construction.

References

R06 Dendrochronology and crossdating U.S. National Park Service

R12 Scientific dating in practice Historic England